WASP-36b

R-0870

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_221222` · created 2026-07-28T23:21:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459935.9571 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.013
Transit depth (Rp/Rs)^2	1.66 +/- 0.33 [%]
Orbital Inclination (inc)	84.09 +/- 1.19
Airmass coefficient 1 (a1)	1.014 +/- 0.011
Airmass coefficient 2 (a2)	-0.009 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#3 - [428, 139]
Optimal Aperture	3.16
Optimal Annulus	6.52
Transit Duration (day)	0.0796 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.013 vs 0.1368 (deviation 0.6σ)
Duration fit vs published	0.0796 d vs 0.0758 d (deviation 0.55σ)
Mid-time O–C	1.3 min
Depth SNR	9.9
Residual scatter	1.05 %

Light curve

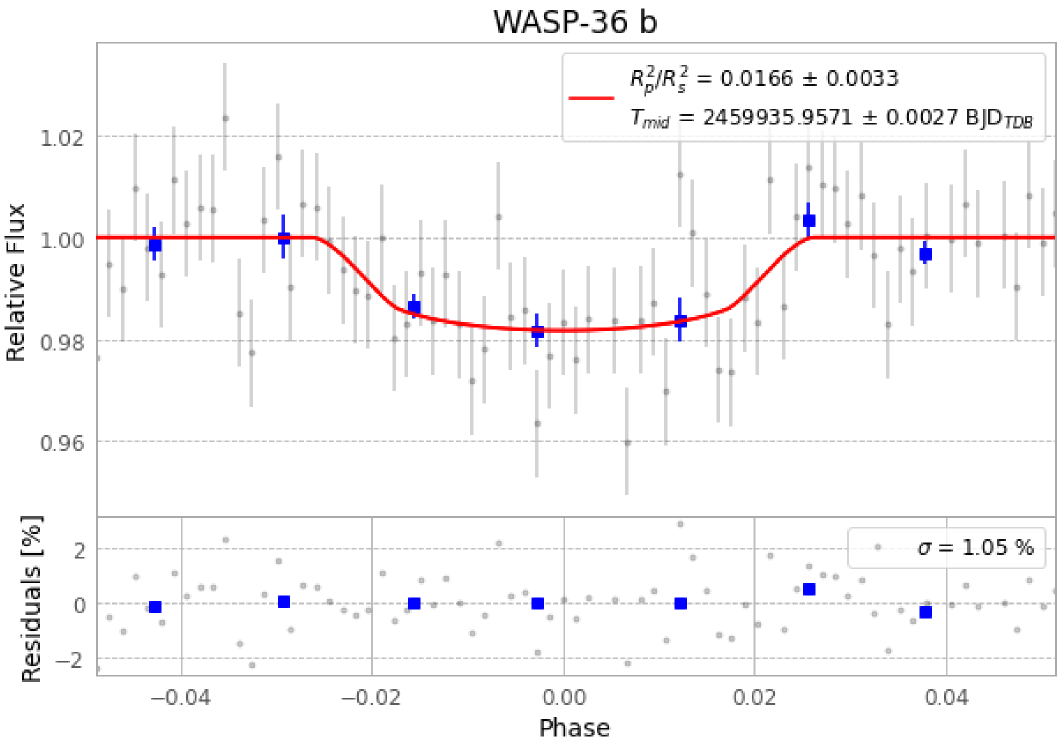

Figure 1 — FinalLightCurve_WASP-36 b_2022-12-22.png (SHA-256 b1efc35bb0392f34…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 171], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9614bb30a93ee94d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

75 FITS frames (49 MB), `WASP-36221222090315.FITS` → `WASP-36221222124515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-221222.log (0e0b048c9041…), FinalLightCurve_WASP-36 b_2022-12-22.png (b1efc35bb039…), FinalLightCurve_WASP-36 b_2022-12-22.csv (77262345598b…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)